

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

TEAM ID	SWTID-2026-1261
PROJECT TITLE	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
MAX MARKS	5 MARKS

CHAPTER – VII

FUNCTIONAL AND PERFORMANCE TESTING

7.1 – Performance Testing & Functional Testing

1. Data Collection

- Gather the housing market dataset from CSV or Excel files.
- Ensure the dataset contains relevant attributes such as Sale Price, Bedrooms, Bathrooms, Square Footage, Lot Area, and Location.

2. Data Loading

- Open Tableau Desktop.
- Click **Connect** → **Microsoft Excel** (or Text File for CSV).
- Select the housing dataset and load it into Tableau.

3. Data Preprocessing

- Check for missing or null values.
- Remove duplicate records if present.
- Correct data types (e.g., Sale Price as Number, Date Sold as Date).
- Rename fields for better readability.

4. Data Exploration

- Analyse the dataset structure.
- Identify key variables affecting housing prices.
- Examine relationships between property features and sale prices.

5. Create Visualizations

- Create bar charts for average sale prices by location.

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- Create line charts to show housing price trends over time.
- Create scatter plots to analyse feature-price relationships.
- Create pie charts or tree maps for property distribution.

6. Apply Filters

- Add filters for:
 - Location
 - Sale Price Range
 - Bedrooms
 - Bathrooms
 - Year of Sale
- Enable interactive filtering for better user analysis.

7. Build Dashboards

- Combine multiple visualizations into dashboards.
- Arrange charts logically.
- Add titles, legends, and descriptive labels.

8. Create Story Points

- Develop Tableau Stories to present key findings.
- Explain important trends and patterns discovered from the data.

9. Testing

Functional Testing

- Verify all charts display correct information.
- Check filter functionality.
- Ensure dashboards respond properly to user interactions.

Performance Testing

- Verify dashboard loading speed.
- Check responsiveness when filters are applied.
- Ensure smooth navigation between dashboard pages.

10. Analysis and Findings

- Identify factors influencing housing prices.
- Compare prices across different locations.
- Analyze the impact of property features on sale prices.

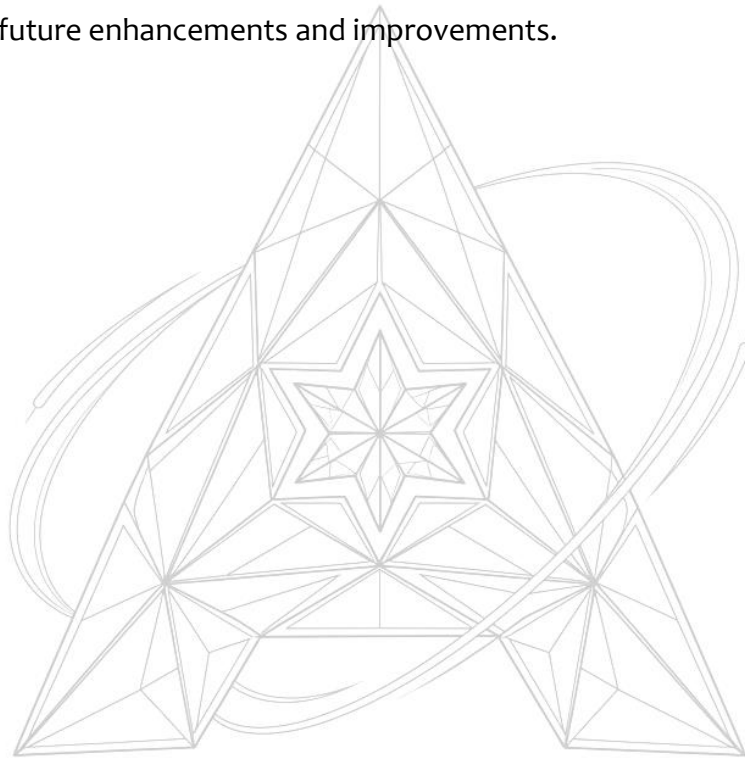
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11. Deployment

- Publish dashboards and stories to Tableau Public.
- Generate shareable links for users.

12. Documentation and Conclusion

- Document methodology, visualizations, and results.
- Summarize key insights obtained from the analysis.
- Suggest future enhancements and improvements.



ALPHA
— INNOVATORS —
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